BUILT — 3D viewer: detailed parametric models (v0.68.0)

The WebGL renderer was already glossy; the geometry was blocky. partgeo.js builds the real part shape from the measured dimensions + FLIS characteristics.

Family-specific meshes

- bolt = hex head + threaded shank · nut = chamfered hex + bore · washer/gasket = ring (gasket with bolt-holes).
- bearing = outer/inner races + a ring of balls · gear = teeth + bore · spring = helix.
- tube = hollow cylinder · o-ring/seal = torus · pin/shaft = chamfered ends · bracket = L + holes · battery = body + posts.

12 families, classified from the item name.

Driven by real measurements

- Pulls overall length / width / height / diameter / bore from the FLIS characteristics already shown in the panel.
- So sizes and proportions match the catalogued part, not a generic block.
- Still honest: a representative parametric model (no CAD geometry exists in a TM), but it looks like the part.

Proportions from the catalogue.

Both renderers, gallery stays fast

- partgeo.js returns a plain {V,F} mesh used by BOTH the WebGL shader (smooth normals, glossy) and the SVG fallback (legacy).
- Gallery thumbnails keep the light primitive (fast grid); opening a part upgrades to the detailed mesh.
- Served at /partgeo.js. No new dependency, offline.

Detail where it counts; speed in the grid.

VERIFIED

partgeo.js node-validated: all 13 families produce finite, non-degenerate meshes (no NaN, valid indices) — bolt 399 verts (hex head + thread grooves), bearing 1342 (races + balls), spring 2120 (helix), gear with teeth. Family classifier 12/12 on real nomenclatures. threed.html wiring + the /partgeo.js route confirmed on host. The same {V,F} reaches the SVG legacy path, so legacy gets the detailed shapes too (flat-shaded).